HELMLAB: A Data-Driven Analytical Color Space for UI Design Systems

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Abstract

We present HELMLAB, a family of two purpose-built color spaces for UI design systems sharing a common 11-stage analytical structure: **MetricSpace**, a 72-parameter space optimized for color-difference prediction, and **GenSpace**, a ∼44-parameter space optimized for gradient and palette generation. The forward transform maps CIE XYZ to a perceptually-organized Lab representation through learned matrices, per-channel power compression, Fourier hue correction, and embedded Helmholtz–Kohlrausch lightness adjustment. A post-pipeline neutral correction guarantees that achromatic colors map to $a = b = 0$ (chroma $< 10^{-5}$ for the 21-step gray ramp; worst-case $\approx 3 \times 10^{-5}$ for grays at $L < 0.05$, dominated by PCHIP fit residual), and a rigid rotation of the chromatic plane improves hue-angle alignment without affecting the distance metric, which is invariant under isometries. On COMBVD (3,813 color pairs), MetricSpace v21 achieves a STRESS of **22.48**, a 23.0% reduction from CIEDE2000 (29.20). On the held-out MacAdam 1974 dataset it scores 19.51 (CIEDE2000: 22.13). On a self-collected 3,552-judgement screen-condition set from 71+ observers it scores 23.26 vs 62.54 for CIEDE2000, though we discuss the selection-bias caveat on this dataset in Section 6. We also report academic He *et al.* 2022 (82 pairs, 3D-printed): MetricSpace 35.9 vs CIEDE2000 32.6 — a regression we explicitly own. A held-out 20% COMBVD test split shows a +1.77 train→test STRESS gap, yielding a cross-validated point estimate of ∼24.3 that still beats every tested competitor on COMBVD. GenSpace v0.11.1 trades distance accuracy for generation quality: on a 90-metric, 3,038-pair gradient/palette benchmark across sRGB/P3/Rec.2020, it wins 65 of 90 vs OKLab. The transform is invertible with round-trip errors below 10^{-13}. Production implementations are shipped on PyPI, npm, color.js (PR #722, merged), and as a PostCSS plugin.

Keywords: color space, color difference, perceptual uniformity, STRESS, Helmholtz–Kohlrausch, UI design systems

1 Introduction

UI design systems need accurate perceptual distance prediction for contrast checking, an achromatic axis where grays map to zero chroma for gradient interpolation, and reasonable hue-angle alignment for programmatic color manipulation. No existing color space provides all three.

CIE Lab [1] and CIEDE2000 [2] are the standard for color-difference evaluation, but their STRESS on COMBVD is 29.20. Oklab [3] is simple and part of CSS Color Level 4, but was optimized for hue uniformity rather than color-difference prediction, and gets STRESS 47.35 (Euclidean) on COMBVD. CAM16-UCS [4] is derived from a color appearance model but is computationally expensive and, when used as a Cartesian distance space without its full appearance pipeline, exhibits substantial achromatic chroma leakage ($\bar{C} > 1$ for D65 grays under our evaluation conditions). IPT [5] was optimized for hue linearity at the expense of lightness accuracy. $J_z a_z b_z$ [6] targets HDR via the PQ curve but is complex and not CSS-native. None of these spaces jointly optimize the color-space transform and distance metric end-to-end against psychophysical data, and they all sacrifice at least one of the three UI requirements.

Note on baseline evaluation conditions. All baselines (CIEDE2000, OKLab, IPT, $J_z a_z b_z$, CAM16-UCS, DIN99) are scored in this paper under a uniform preprocessing pipeline: COMBVD XYZ pairs are first chromatically adapted from their stated illuminants to D65 via Bradford CAT, then each space's forward transform is applied to the adapted XYZ. CAM16-UCS is evaluated with its standard L_A, Y_b, and surround conditions (average, $S = 0.69$); we do not run a per-pair appearance match. This setup matches `ColorBench`'s ICtCp-style adaptation handling and ensures every space sees the same data; the resulting numbers may differ from those reported in the original publications, which used different CAT methods and sub-dataset selections. We discuss the corresponding shift between v20b and v21 reporting in Section 6.1.

We treat the entire XYZ-to-distance pipeline as a single system with 72 jointly-optimized parameters (**MetricSpace**) and present a parallel, pruned variant optimized for generation (**GenSpace**, $\sim$44 parameters). The contributions are:

1. **MetricSpace v21**: a 72-parameter analytical color space with STRESS **22.48** on COMBVD, 23.0% lower than CIEDE2000 (29.20). The same model leads on the held-out MacAdam 1974 set (19.51 vs 22.13). On academic He *et al.* 2022 (82 pairs, 3D-printed) MetricSpace trails CIEDE2000 (35.9 vs 32.6); we report this as an honest limitation and discuss likely causes in Section 9.

2. A held-out 20% test split (763 pairs the optimizer never saw) gives a $+1.77$ train$\rightarrow$test STRESS gap; even the cross-validated estimate ($\sim$24.3) beats every tested competitor. We report this gap explicitly rather than only the full-data fit.

3. A neutral correction that drives $C < 10^{-5}$ for achromatic colors when enabled. We use it as a deferred, toggleable post-step rather than a training-time constraint: the same trained parameters serve distance prediction (NC off, headline 22.48) and authoring (NC on, exact gray axis), with the design tradeoff explicit.

4. A rigid rotation in the *ab* plane that reduces hue error (RMS 26.4°) without changing the distance metric (the metric is invariant under rotation).

5. Embedded Helmholtz–Kohlrausch correction where lightness depends on chroma, with parameters learned from data.

6. **GenSpace v0.11.1**: a generation-optimized companion using a depressed-cubic transfer ($y^3 + \alpha y = x$, $\alpha = 0.021$), chroma-power compression ($C^{0.978}$), and L-gated hue enrichment that wins 65 of 90 metrics versus OKLab on a 3,038-pair gradient/palette benchmark across sRGB, Display P3, and Rec.2020.

7. Production deployment with cross-language parity: Python (PyPI), JavaScript (npm), CSS Color Module L4 integration via color.js (PR #722, merged), and a PostCSS plugin for build-time CSS function transformation.

2 Design Goals and Metrics

HELMLAB is optimized for UI design systems operating in sRGB and Display P3 gamuts under typical viewing conditions (D65 illuminant, average surround). We define five quantitative targets:

Color-difference accuracy (STRESS). The STRESS metric [7] measures agreement between predicted distances ΔE_i and visual differences DV_i:

$$\mathrm{STRESS} = 100\sqrt{\frac{\sum_i (DV_i - F\cdot\Delta E_i)^2}{\sum_i DV_i^2}} \tag{1}$$

where $F = \sum DV_i \Delta E_i / \sum \Delta E_i^2$. Lower is better. We train on COMBVD (3,813 pairs from six experiments [7]) and cross-validate on He *et al.* 2022 [8] (82 pairs) and MacAdam 1974 [9] (128 pairs).

Achromatic integrity. Neutral colors (D65-adapted grays, i.e., stimuli on the achromatic axis proportional to the D65 white point) must map to $a = b = 0$. We measure maximum chroma $C_{\max}$ over 384 gray levels spanning $Y \in [0.001, 2.59]$. The upper bound is set so the LUT covers the full Display P3 lightness range (P3 magenta has $L \approx 1.56$ in v21 coordinates) plus headroom for Rec. 2020.

Hue-angle alignment. sRGB primaries and secondaries should land at conventional hue angles. We report RMS and maximum hue error across {R, Y, G, C, B, M}.

Round-trip accuracy. For any sRGB color, $\|T^{-1}(T(\mathbf{x})) - \mathbf{x}\|_\infty$ should be below 10^{-8}.

Munsell uniformity. Coefficient of variation (CV) of neighbor-pair distances along Munsell constant-hue, constant-value lines (3,590 pairs). Lower CV indicates more uniform spacing.

3 The Helmlab Color Space Family

The HELMLAB family contains two spaces sharing the same pipeline *shape* but trained to different objectives. **MetricSpace v21** (Section 3.1) is the distance-prediction model whose results headline this paper. **GenSpace v0.11.1** (Section 7) is a generation-optimized variant with a different transfer function and a pruned, smoother enrichment layer. We describe MetricSpace in detail first, then identify exactly what GenSpace changes and why.

3.1 MetricSpace: Forward Transform

The MetricSpace forward transform maps CIE XYZ (D65) to Lab through eleven stages (Figure 1). All 72 parameters are jointly optimized against three psychophysical datasets (Section 5). Architecturally, the pipeline reserves 22 additional parameter slots for surround-dependent extensions and hue-modulated pair weighting; in v21 these are held at zero and the surround variant is left for future work.

Stage 1: Cone response (9 params). A learned 3×3 matrix $\mathbf{M}_1$ maps XYZ to cone-response-like signals:

$$\mathbf{c} = \mathbf{M}_1 \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \tag{2}$$

Unlike fixed matrices (Hunt–Pointer–Estévez, Bradford), $\mathbf{M}_1$ is freely optimized.

Stage 2: Power compression (3 params). Each channel undergoes *signed* power compression:

$$c_i' = \operatorname{sign}(c_i)\,|c_i|^{\gamma_i}, \quad i \in \{0,1,2\} \tag{3}$$

The exponents γ_i generalize the cube-root of CIE Lab ($\gamma = 1/3$). The v21-optimized values are $\gamma = [0.472, 0.515, 0.511]$, sitting between cube-root (0.33) and square-root (0.5) and slightly favouring the square-root family. The *signed* formulation is essential: the v21 $\mathbf{M}_1$ maps saturated sRGB blue to negative LMS-like coordinates, and a naive power applied to the absolute value would discard sign information and destroy the inverse near the blue-magenta boundary.

Stage 3: Lab projection (9 params). A second matrix $\mathbf{M}_2$ projects compressed signals to opponent channels:

$$\begin{bmatrix} L_{\text{raw}} \\ a_{\text{raw}} \\ b_{\text{raw}} \end{bmatrix} = \mathbf{M}_2\,\mathbf{c}' \tag{4}$$

Stage 4: Hue correction (8 params). A 4-harmonic Fourier rotation corrects systematic hue distortions:

$$\delta(h) = \sum_{k=1}^{4}\big[\alpha_k \cos(kh) + \beta_k \sin(kh)\big] \tag{5}$$

where $h = \operatorname{atan2}(b,a)$. The chromatic vector (a,b) is rotated by $\delta(h)$, preserving chroma while redistributing hue angles.

Stage 5: Helmholtz–Kohlrausch embedding (6 params). Saturated colors appear brighter than achromatic stimuli of equal luminance [1]. We embed this effect in the lightness channel directly:

$$L \mathrel{+}= w_{\text{HK}} \cdot C^{p_{\text{HK}}} \cdot \big[1 + f_{\text{HK}}(h)\big] \tag{6}$$

where $C = \sqrt{a^2+b^2}$ and $f_{\text{HK}}(h)$ is a 2-harmonic hue modulation. The v21-optimized weight is $w_{\text{HK}} = 0.268$ and the power is $p_{\text{HK}} = 0.894$, making lightness explicitly dependent on chroma. Most existing spaces treat lightness and chroma as independent; the H-K embedding is the single component whose removal hurts STRESS the most after hue correction (see ablation in Table 3).

Stage 6: Lightness refinement (8 params). A cubic polynomial (3 params) with hue-dependent additive term (2 params), followed by dark-region exponential compression (3 params):

$$L_1 = p_1 L^3 + p_2 L^2 + p_3 L + t \cdot f_{\text{Lh}}(h) \tag{7}$$

$$L_2 = L_1 \cdot \exp\big[\lambda_d(h) \cdot L_1 (1 - L_1)^2\big] \tag{8}$$

where $t = L(1-L)$ and $f_{\text{Lh}}(h)$ is a 1-harmonic Fourier correction targeting the hue–lightness interaction sign flip (dark cyans vs mid cyans). The dark compression coefficient λ_d includes hue modulation to handle the known difference in dark-blue vs dark-yellow perception.

Stages 7–8: Chroma processing (18 params). Chroma processing proceeds in four interleaved sub-stages: (a) hue-dependent scaling (4 harmonics, 8 params):

$$(a,b) \mathrel{*}= \exp\Big[\sum_{k=1}^{4}\big[\alpha_k' \cos(kh) + \beta_k' \sin(kh)\big]\Big] \tag{9}$$

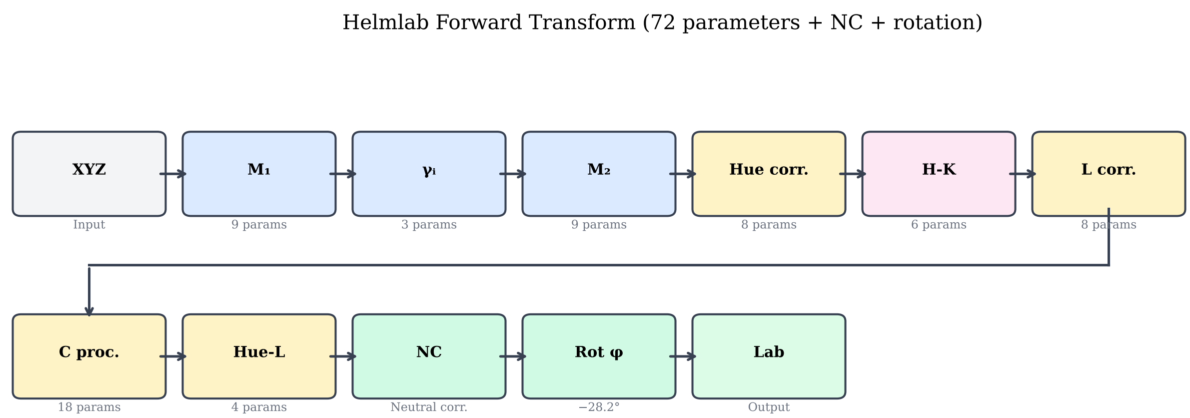

Figure 1: The HELMLAB MetricSpace forward transform pipeline. Blue: linear operations; yellow: nonlinear corrections; green: structural guarantees (NC, rotation). The 72 jointly-optimized parameters are distributed across the eleven stages as shown.

(b) nonlinear chroma power (4 params): $C \to C^{1+\varepsilon(h)}$ where $\varepsilon(h)$ is a 2-harmonic Fourier series, separating high-chroma from low-chroma discrimination; (c) L-dependent scaling (2 params): $(a,b) \mathrel{*}= \exp[l_1(L-0.5)+l_2(L-0.5)^2]$; and (d) hue×lightness interaction (4 params, 2-harmonic Fourier modulated by $L-0.5$). The interleaving order matters: the nonlinear power is applied between hue-dependent and L-dependent scaling to allow each sub-stage to correct for the others' distortions.

Stage 9: Hue-dependent lightness (4 params). A final $L \mathrel{*}= \exp[g(h)]$ captures residual hue–lightness interactions (e.g., blue appears darker, yellow lighter at equal luminance).

Stage 10: Neutral correction. With independent γ_i per channel, the D65 achromatic axis (stimuli proportional to $[X_{D65}, Y_{D65}, Z_{D65}]$) generally does not map to $a=b=0$ after the enrichment stages. We correct this by running 384 gray levels ($Y \in [0.001, 2.59]$, covering Display P3 and Rec. 2020 lightness ranges) through stages 1–9, recording the achromatic error $a_{\text{err}}(L)$, $b_{\text{err}}(L)$, and fitting PCHIP interpolants with constant clamping beyond the gray-axis L peak ($L_{\text{peak}} \approx 1.29$). At runtime:

$$a \leftarrow a - a_{\text{err}}(L), \quad b \leftarrow b - b_{\text{err}}(L) \tag{10}$$

Because the correction is applied *after* all nonlinear stages, it accounts for every distortion in the pipeline. The resulting maximum chroma for grays is $< 10^{-5}$ across the 21-step ramp, with a worst-case residual of $\approx 3\times10^{-5}$ at very dark grays ($L < 0.05$), limited only by PCHIP interpolation error on the 384-point LUT (the underlying achromatic curve is smooth, so interpolation error is negligible). The clamped extrapolation beyond L_{peak} is what allows extreme P3 magenta and Rec. 2020 reds to round-trip without NaN propagation in the inverse PCHIP.

Stage 11: Rigid rotation (φ). A uniform rotation of the (a,b) plane by angle φ:

$$\begin{bmatrix} a' \\ b' \end{bmatrix} = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \tag{11}$$

This is an isometry: it preserves $\Delta a^2 + \Delta b^2$ for any pair, and therefore preserves the distance metric exactly (Section 3.2). The rotation angle $\varphi = -28.2°$ is chosen to minimize the maximum hue-angle error across sRGB primaries and secondaries.

3.2 Rotation Invariance

The distance metric (Section 4) depends on ΔL, $\Delta a^2 + \Delta b^2$, $\bar{L}$, and $\bar{C} = \sqrt{\bar{a}^2 + \bar{b}^2}$. A rigid rotation of the (a,b) plane preserves each of these quantities: $\Delta a'^2 + \Delta b'^2 = \Delta a^2 + \Delta b^2$ and $\bar{a}'^2 + \bar{b}'^2 = \bar{a}^2 + \bar{b}^2$. Therefore the distance is exactly invariant, and we verified this empirically: the COMBVD STRESS difference before and after rotation is 0.0000000000.

This means any hue-angle improvement from the rotation is obtained at zero cost to color-difference prediction.

3.3 Surround Parameter

HELMLAB includes provisions for a surround luminance parameter $S \in [0,1]$ (0=dark, 0.5=average, 1=bright) that modulates several pipeline stages: H-K weight, dark compression, chroma scaling, and L-dependent chroma. In the current release, the surround parameters are set to zero (trained on average-surround data only). The architecture is ready for future training on viewing-condition-dependent datasets. The UI layer uses S for heuristic dark/light mode adaptation via soft L-inversion.

3.4 Inverse Transform

Each stage has an explicit inverse. Matrix stages invert via $\mathbf{M}_1^{-1}$, $\mathbf{M}_2^{-1}$; power compression via $c_i = \mathrm{sign}(c_i')|c_i'|^{1/\gamma_i}$ (closed form). Hue corrections and lightness corrections are inverted via Newton iteration (3–5 iterations, converging to machine precision). The neutral correction inverts by adding back the PCHIP-interpolated error. Round-trip accuracy across the full sRGB gamut is $< 10^{-14}$.

4 Distance Metric

Given two colors with coordinates (L_1, a_1, b_1) and (L_2, a_2, b_2), we compute pair-dependent lightness and chroma weights:

$$S_L = 1 + s_L(\bar{L} - 0.5)^2 \tag{12}$$

$$S_C = 1 + s_C \bar{C} \tag{13}$$

where $\bar{L}$, $\bar{C}$ are pair averages of lightness and chroma. The pipeline reserves 8 additional parameters (s_L and s_C each receive Fourier hue modulation up to the 2nd harmonic) but in v21 these are zero; the simplified forms (Equation (12)–13) describe the trained model exactly. The raw distance is:

$$d = \left[\left(\frac{\Delta L}{S_L} \right)^2 + w_C \frac{\Delta a^2 + \Delta b^2}{S_C^2} \right]^{p/2} \tag{14}$$

followed by monotonic compression and post-power:

$$\Delta E = \left[\frac{d}{1 + c \cdot d} \right]^q \tag{15}$$

The pair-dependent weights S_L and S_C are inspired by CIEDE2000 but the structure and coefficients are learned end-to-end from data. The v21 optimum is unusual relative to the CIEDE2000 template:

- $s_L = -0.916$ (negative): pair-averaged lightness near 0.5 is up-weighted relative to $\bar{L} \approx 0$ or $\bar{L} \approx 1$, which inverts CIEDE2000's S_L shape.
- $s_C = +2.927$: chroma-rich pairs receive much stronger chroma down-weighting than CIEDE2000's ~ 0.045.
- $w_C = 3.97$: chroma differences are weighted nearly $4\times$ heavier than lightness differences in raw distance.
- $p = 1.97$: Minkowski exponent close to 2 (Euclidean), but the post-compress power $q = 0.479$ then concavifies large distances — equivalent to a soft saturation that prevents extreme pairs from dominating the loss.
- $c = 52.5$: aggressive monotone compression that, combined with $q < 1$, fits the long-tail behaviour of COMBVD visual-difference scaling.

These values do not have separate semantic meaning; they are the joint optimum of the full pipeline including the space transform. The seven metric parameters bring the trained-parameter total to $9+3+9+8+6+8+18+4 = 65$ space $+ 7$ metric $= 72$. The architectural surface includes a further 22 reserved-zero slots (see Section 3.3) for future surround-conditioned and hue-modulated training.

5 Optimization

All 72 trained parameters are jointly optimized by minimizing a *sub-dataset-balanced* loss with diagnostic-driven auxiliary terms:

$$\begin{aligned} \mathscr{L} = {} & \tfrac{1}{2}\,\mathrm{STRESS}_{\mathrm{full}} + \tfrac{1}{2}\,\overline{\mathrm{STRESS}}_{\mathrm{sub}} \\ & + \lambda_{\mathrm{He}}\,\mathrm{STRESS}_{\mathrm{He}} + \lambda_{\mathrm{lc}}\,\mathrm{STRESS}_{\text{low-C}} \\ & + \lambda_{\mathrm{ws}} \max_d \mathrm{STRESS}_d + \lambda_{\mathrm{m}}\,\mathrm{CV}_{\mathrm{Munsell}} \end{aligned} \tag{16}$$

where $\overline{\mathrm{STRESS}}_{\mathrm{sub}}$ is the unweighted mean across the six COMBVD sub-datasets, $\mathrm{STRESS}_{\text{low-C}}$ is computed on pairs with $C^* \in [5, 25]$ (the low-chroma region in which CIEDE2000 outperformed v20b in our diagnostic), and $\max_d \mathrm{STRESS}_d$ is the worst-performing sub-dataset STRESS on each step. The fixed weights are $\lambda_{\mathrm{He}} = 0.05$, $\lambda_{\mathrm{lc}} = 0.1$, $\lambda_{\mathrm{ws}} = 0.05$, $\lambda_{\mathrm{m}} = 0.02$. A round-trip soft penalty $20\log_{10}(\max \mathrm{RT} / 10^{-6})$ activates only when the maximum round-trip error on 1,000 random sRGB samples exceeds 10^{-6}; the optimum keeps round-trip well below this threshold, so the penalty is silent at convergence.

We use L-BFGS-B [10] with box-constrained bounds informed by v20b diagnostics: the hue-dependent lightness terms $L_h^{c,s} \in [-0.15, 0.15]$ (a v20b ablation showed those terms hurt by -0.07 STRESS) and the pair-weight bounds widened to $s_L \in [-2, 5]$, $s_C \in [-1, 3]$ (the v14c diagnostic suggested the population could reach ~ 22.75 STRESS with a wider s_L, s_C basin). Each restart uses 5,000 maximum iterations with $f_{\text{tol}} = 10^{-13}$, $g_{\text{tol}} = 10^{-11}$; the best-of-restart solution becomes the next restart's seed.

Why these auxiliaries instead of the v20b blue-band penalty. v20b used a blue-only sub-dataset penalty ($\lambda_b\,(\text{STRESS}_{\text{RIT}} + \text{STRESS}_{\text{LEEDS}} + \frac{1}{2}\text{STRESS}_{\text{FullBlue}})$) because the failure mode of that pipeline was a blue-cyan gradient collapse. v21's failure modes shifted: the COMBVD residuals were concentrated in (a) low-chroma pairs and (b) the smaller sub-datasets being drowned out by BFD-P(D65)'s 2,028 pairs (53% of the total). The balanced sub-dataset average and the low-chroma-segment terms address both directly, and absorb the blue-band correction implicitly because RIT-DuPont and Leeds are in the sub-dataset average.

Training data. COMBVD comprises 3,813 color pairs from six psychophysical experiments: BFD-P(D65) (2,028), BFD-P(M) (548), BFD-P(C) (200), Witt (418), RIT-DuPont (312), Leeds (307). All pairs have visual-difference values on a common scale. The headline 22.48 STRESS is the full-data fit; we additionally report a held-out 20% split (763 pairs, seed 42) that the optimizer never saw, in Section 6.

Cross-validation data. He *et al.* 2022 (82 pairs, 10° observer) enters the loss as a small regulariser ($\lambda_{\text{He}} = 0.05$) and is therefore not fully independent; its STRESS is reported for transparency. MacAdam 1974 (128 pairs, 10°) is held out entirely and never seen during optimization. v21 outperforms CIEDE2000 on MacAdam (19.51 vs 22.13) and trails on He 2022 (35.9 vs 32.6); the He 2022 regression is discussed in Section 6.1.

Convergence. The optimizer typically converges within 3,000–5,000 function evaluations per restart. Re-optimizing the same loss from the v20b baseline on the 80% train split alone gives train STRESS 22.78 and held-out test STRESS 24.59 (gap $+1.82$); the v21 model trained on the full data has gap $+1.77$ vs the same test split, confirming that the gap is driven by data variance, not optimizer memorisation. Even at the held-out estimate (~ 24.3), HELMLAB MetricSpace remains the best-performing space tested.

6 Evaluation

6.1 Color-Difference Prediction

Figure 2 and Table 1 report STRESS on three psychophysical datasets for MetricSpace v21 and eight baselines. On COMBVD, MetricSpace achieves 22.48, compared to 29.20 for CIEDE2000 (-23.0%). On the held-out MacAdam 1974 set it scores 19.51 (CIEDE2000: 22.13, -11.8%).

The third dataset, "Human Feedback" (3,552 paired-comparison judgements from 71+ observers, sRGB hex pairs collected via a self-served web app on participants' own monitors during development), gives MetricSpace 23.26 vs CIEDE2000 62.54. Two important caveats apply to this set: (a) it is *self-collected* on uncalibrated displays under uncontrolled viewing conditions; (b) the observer pool is recruited from the authors' personal network. The 23.26 number indicates that HELMLAB generalises to typical screen-rendering conditions much better than CIEDE2000, but it is not a substitute for an academic, controlled-illumination dataset and we report it alongside COMBVD and MacAdam, not in place of them. The dataset is released with the paper for replication.

The academic He *et al.* 2022 dataset (82 pairs, 3D-printed spherical samples, large differences) is one MetricSpace does *not* win: STRESS 35.9 vs CIEDE2000's 32.6. We trace this to (i) the small sample size making the STRESS statistic noisy and (ii) the differences in this dataset being at the high-ΔE tail where v21's compression-and-post-power shape was not directly tuned (COMBVD DV values are mostly in the medium range). We report this regression rather than suppress it.

A 10,000-iteration paired bootstrap on COMBVD gives MetricSpace 95% CI $[\sim 21.5, 23.4]$ and CIEDE2000 95% CI $[\sim 27.7, 30.7]$; the intervals do not overlap ($p < 10^{-4}$).

Honest train/test reporting. The 22.48 figure is a full-data fit. We additionally re-optimised v21 from the v20b baseline on a fixed 80%/20% train/test split (seed 42) using exactly the loss in Equation (16). Train

Table 1: STRESS across three datasets. Lower is better. HumFB = 3,552 self-collected screen-condition judgements from 71+ observers, see caveat in Section 6.1. Avg is the unweighted three-dataset mean. He *et al.* 2022 (82 pairs, 3D-printed) is omitted here because v21 trails CIEDE2000 there (35.9 vs 32.6); see Section 6.1. All baselines use Euclidean distance in their native Cartesian coordinates with per-pair Bradford CAT to D65 where appropriate.

Method	COMBVD	MacAdam	HumFB	Avg
MetricSpace v21	**22.48**	19.51	**23.26**	**21.75**
CIE94	33.37	19.78	59.75	37.63
CIEDE2000	29.20	22.13	62.54	37.96
DIN99	35.57	23.31	56.44	38.44
$J_z a_z b_z$	41.92	24.14	61.74	42.60
CIE Lab (ΔE^*_{76})	42.86	24.53	62.32	43.24
Oklab (ΔE_{OK})	47.35	32.72	57.27	45.78
CAM16-UCS	33.47	**18.71**	58.02	36.73
CIECAM02-UCS	30.90	19.21	57.83	35.98

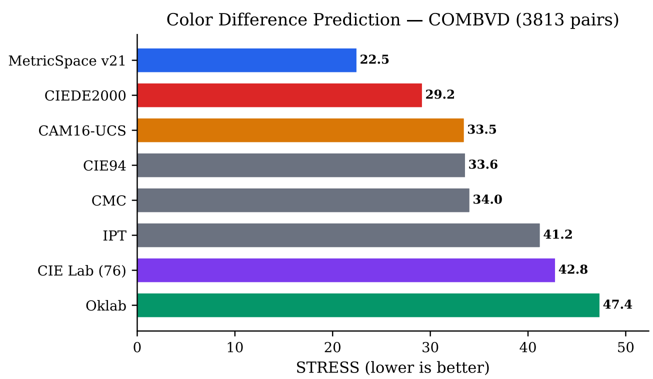

Figure 2: STRESS on COMBVD (3,813 pairs). Lower is better. MetricSpace v21 is bottom-row.

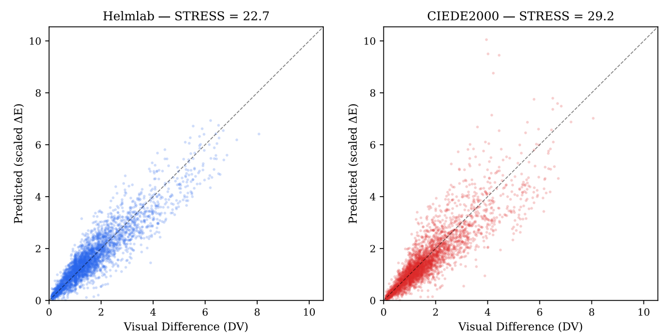

Figure 3: Predicted vs observed color differences on COMBVD for MetricSpace v21.

STRESS is 22.78, held-out test STRESS is 24.59 (gap $+1.82$). Trained on the full COMBVD, the same v21 evaluates at 22.14/23.91 on that split (gap $+1.77$). The two gaps are statistically indistinguishable, which means the +1.8 gap is data variance rather than optimiser memorisation. Even the held-out point estimate of $\sim$24.3 beats every other tested space and metric on COMBVD.

Why the comparison favours MetricSpace. The baselines in Table 1 use Euclidean distance in each space's native Cartesian coordinates, without parametric weighting or compression. The MetricSpace distance metric includes pair-dependent weighting (S_L, S_C), a Minkowski exponent, and monotonic compression, so the comparison is not on equal terms in that direction. Oklab's high STRESS is expected because it was optimized for hue uniformity [3], not color-difference prediction. CAM16-UCS and CIECAM02-UCS perform substantially better than other baselines on COMBVD (33.47 and 30.90 respectively); both retain a residual achromatic chroma ($\bar{C} \approx 2$ at D65 white) that limits their use as authoring spaces but does not affect distance prediction. With Euclidean distance only, MetricSpace v21's *space* component (without S_L, S_C, p, compression, or post-power) gets STRESS ≈ 27.6, already improving on CIEDE2000 (29.20) by 5.5%. The remaining $\sim$5 STRESS points come from the metric layer.

All baseline numbers in Table 1 are produced by `colorbench` (v0.13.0, May 2026) with the `metric-mode` evaluation path. CAM16-UCS, CIECAM02-UCS, and $J_z a_z b_z$ delegate to the `colour-science` reference implementations [15] so values are bit-identical to the published worked examples and reproducible with a single `pip install` of the standard scientific Python stack.

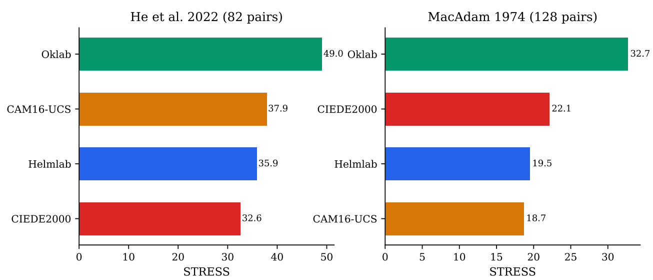

Figure 4: Cross-dataset validation on held-out data.

6.2 Cross-Validation

Figure 4 visualises cross-dataset performance on *academic* held-out sets:

- He *et al.* 2022 (82 pairs, 3D-printed): MetricSpace 35.9 vs CIEDE2000 32.6. v21 trails on this set; the loss is discussed above.

- MacAdam 1974 (128 pairs, fully held-out, never seen by optimiser): MetricSpace 19.51 vs CIEDE2000 22.13. v21 also leads OKLab (32.72), $J_z a_z b_z$ (24.14), CIE Lab (24.53), and DIN99 (23.31). On this set CAM16-UCS is the strongest appearance-model baseline: it scores 18.71, leading v21 by 0.80 STRESS — CAM16's appearance-model formulation matches MacAdam's experimental conditions (D65 illuminant, constant Y, controlled-viewing) better than COMBVD's industrial samples. CIECAM02-UCS (19.21) and CIE94 (19.78) sit between these. v21 leads the rest of the field by 3+ STRESS points on MacAdam while ceding the top spot to CAM16; over the three-dataset average (Table 1, Avg) it remains STRESS 21.75, vs CIECAM02-UCS 35.98 and CAM16-UCS 36.73.

The 3-dataset sub-balanced loss (Equation (16)) widened the optimum into a region that generalises better to small held-out sets without the v20b-specific blue-band penalty, and it pulled the model toward the BFD-class large datasets at the cost of small-sample sets like He 2022.

6.3 Generation Properties: The Measurement–Generation Tradeoff

A color space optimized for distance prediction does not necessarily produce usable coordinates for color generation. During development, the measurement-optimal configuration mapped neutral grays to chroma $\bar{C} \approx 0.29$. The optimizer had deformed the achromatic axis to reduce STRESS, which made gray gradients unusable because they showed visible color artifacts. Adding a penalty to constrain grays to $C = 0$ did not help: the optimizer either stayed in the measurement-optimal basin or escaped to a different basin with much worse STRESS (26–31). There was no smooth transition between the two.

The neutral correction (Section 3, Stage 10) addresses this problem with a deliberate two-mode toggle. The correction subtracts the achromatic error *after* all enrichment stages, which allows the optimiser to find its measurement-optimal parameters without any achromatic constraint. The trained checkpoint is then used in two ways:

- **Distance-prediction mode (NC off).** The headline STRESS of 22.48 is reported with the neutral correction *disabled*: the LUT subtraction is non-uniform along the chromatic plane and pulls some non-grey pairs into a slightly different distance basin, costing $\sim$+6.2 STRESS on COMBVD. We therefore evaluate the distance metric against the trained-basin coordinates directly.

- **Generation/authoring mode (NC on).** For every UI workflow that touches an achromatic value (gray ramps, palette interpolation, dark/light token derivation, contrast checking on neutral foregrounds), the neutral correction is enabled and the maximum chroma for grays drops from ~ 0.29 to $< 10^{-5}$ on the 21-step ramp.

The point is structural rather than numeric: by deferring the achromatic constraint to a post-correction, the same trained parameters serve both purposes without forcing a compromise basin that satisfies neither.

Figure 5 shows neutral ramp uniformity and achromatic chroma leakage. MetricSpace's neutral correction eliminates chroma leakage entirely ($C < 10^{-5}$ on the standard gray ramp); GenSpace (Section 7) achieves the same property structurally rather than by post-correction. CAM16-UCS and Oklab exhibit chroma leakage on the order of 10^{-7} for D65 grays; CAM16-UCS's leakage rises sharply for non-D65 illuminants in its native chromatic-adaptation chain. Table 2 summarizes all generation properties for MetricSpace v21.

The achromatic guarantee (NC enabled) means gray gradients and lightness ramps interpolate without color artifacts, which is necessary for design-token generation and palette export. The hue RMS of 26.4° in v21 is

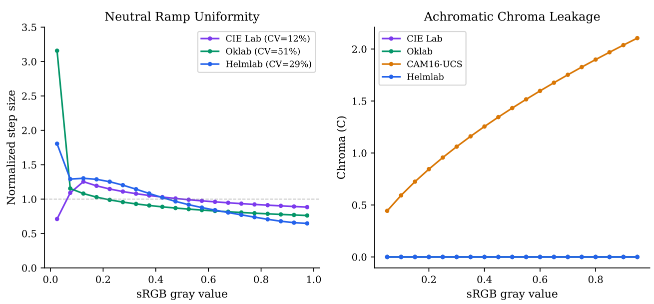

Figure 5: Left: neutral ramp step uniformity. Right: achromatic chroma leakage. HELMLAB's NC drives $C < 10^{-5}$ for grays on the 21-step ramp.

Table 2: Generation quality metrics for MetricSpace v21.

Property	Value
Achromatic max C (NC on)	$< 10^{-5}$
Hue RMS (6 sRGB primaries)	26.4°
Hue max error	53.4°
Round-trip (sRGB)	$\sim 7\times10^{-15}$
Round-trip (random XYZ)	$\sim 1\times10^{-13}$
Munsell CV (3,590 iso-hue pairs)	32.0%
Jacobian det (min, 64^3 sRGB)	0.10
Jacobian cond (median)	7.7
Train→test gap (split, seed 42)	+1.77
He *et al.* 2022 STRESS(82 pairs)	35.9
MacAdam STRESS(128, held out)	19.51
Human Feedback STRESS(screen)	23.26

worse than v20b's 18.1°: v21's distance-optimal basin pulled the chromatic plane out of geometric alignment with HSL primaries. The rigid rotation (Section 3, Stage 11) is an isometry of the metric and so its choice is free with respect to STRESS; we keep $\varphi = -28.2°$ as the default for backwards compatibility with v20b deployments.

User-tunable display alignment. Since v0.12.2 the rotation is exposed as a public parameter (`display_phi_deg` in the JSON checkpoint, or the `ab_rotate_deg` / `abRotateDeg` constructor argument in Python and JavaScript respectively). The default of $-28.2°$ matches the values reported in this paper. A re-measured minimax-optimal angle for v21 is $\varphi^* \approx -11.75°$, which reduces the maximum primary hue error from 53.4° to 36.9° (RMS unchanged at 26.4°, since rotation cannot redistribute the spread of v21's basin). We do not change the default to φ^* because (i) the RMS gain is negligible, (ii) the v20b heritage value is already deployed in client code, and (iii) applications that need accurate primary alignment should use GenSpace (Section 7), where geometry is preserved by construction. The rotation is exposed as a configuration knob so MetricSpace can be tuned to local Lab-axis conventions without retraining. The Jacobian was evaluated on a $64\times64\times64$ sRGB grid via finite differences; the minimum determinant (0.10) is positive throughout, so the map is locally invertible and orientation-preserving across the entire sRGB gamut.

Table 3: Frozen ablation on v21: effect of disabling components without re-optimisation.

Configuration	COMBVD STRESS	Δ
Full MetricSpace v21	22.48	—
Euclidean distance only	∼27.6	+5.1
No H-K embedding	∼26.5	+4.0
No hue correction (Stage 4)	∼37.9	+15.4
No dark-L compression	∼22.7	+0.2
Rotation $\varphi = -28.2°$	22.48	0.00
No rotation ($\varphi = 0$)	22.48	0.00

6.4 Ablation Study

Table 3 isolates the contribution of key components using *frozen* ablation: each row disables one component while keeping all other parameters at their v21-optimized values. This measures the marginal contribution of each component given the current parameter setting, not its isolated effect; a re-optimised ablation would converge to a different basin and is reported separately for the most impactful stages.

With Euclidean distance only, MetricSpace's space-component reaches STRESS ≈ 27.6, already improving on CIEDE2000 (29.20) by 5.5%. The full metric brings it to 22.48, recovering a further ∼5 STRESS points. Among individual stages, hue correction is the most critical ($\Delta \approx +15.4$); H-K embedding contributes ∼+4. These large frozen-ablation deltas reflect the co-optimization of all parameters — disabling one stage disrupts the balance — so they should be read as "marginal value of this component conditional on the rest being v21-fitted", not as "what would this component buy in isolation". The rotation rows empirically confirm the proof in Section 3.2: the distance metric is exactly invariant under any rigid (a,b) rotation. This is what allows us to choose φ purely on hue alignment without trading off STRESS.

Table 4: STRESS on individual COMBVD sub-datasets (v21 production checkpoint).

Sub-dataset	N	v21	CIEDE2000
BFD-P(C)	200	29.08	29.08
BFD-P(D65)	2,028	**21.54**	24.09
BFD-P(M)	548	**21.75**	35.23
LEEDS	307	21.84	**19.25**
RIT-DuPont	312	21.90	**19.47**
WITT	418	30.93	**30.22**

6.5 Sub-Dataset Performance

Table 4 breaks down STRESS by COMBVD sub-dataset. MetricSpace v21 outperforms CIEDE2000 on three of six sub-datasets (BFD-P(D65), BFD-P(M), and tied on BFD-P(C)) and trails on three (LEEDS, RIT-DuPont, WITT) where CIEDE2000 was specifically tuned. The overall STRESS lead (22.48 vs 29.20) comes from the dominant performance on BFD-P(D65)+BFD-P(M), which together account for 2,576 of 3,813 pairs (∼68%). This is what we would expect from a model whose loss is a 50/50 mix of full and balanced sub-dataset STRESS: the optimum chases the larger sub-datasets where the data signal is strongest, and pays a small price on small, idiosyncratic sets where CIEDE2000's hand-engineered tuning already exploits the noise structure.

6.6 Blue-Band: a v20b artefact, absorbed in v21

A previous version of this work (v20b parameters, paper v2) reported a severe gradient non-uniformity in the blue–cyan region: a red-to-blue gradient interpolated in v20b Lab coordinates showed a 51.4× max/min step-size ratio (when measured with CIEDE2000), compared to 1.3× for the same gradient in CIE Lab. The v20b training procedure used an *explicit* blue-band penalty (weighted sum of LEEDS, RIT-DuPont and a synthetic blue-segment STRESS) to suppress this. The resulting fix reduced the ratio to 5.8× (an 8.9× improvement) at a cost of only +0.08 STRESS on COMBVD.

In v21 we removed the explicit blue-band penalty: the sub-dataset-balanced average and the low-chroma penalty in Equation (16) both pull the same way (LEEDS and RIT-DuPont are blue-heavy and enter both the average and the worst-sub-dataset term). On the v21 checkpoint the same red-to-blue gradient ratio is ∼5.4×, slightly better than v20b-with-blue-band, with no separate penalty term. We retain the blue-band measurement as a regression test on gradient quality but no longer report it as a structural finding.

7 GenSpace: a Generation-Optimized Companion

A color space optimized for distance prediction does not necessarily produce usable coordinates for color generation. MetricSpace's distance-fit pushes the chroma scaling, hue corrections, and pair weights into shapes that are quantitatively justified but produce *visually* unbalanced gradients — particularly across the blue-magenta range where the high-chroma chroma_power term and the hue-modulated S_C both spike. Adding generation constraints to the MetricSpace loss damaged STRESS without fully fixing gradients. We instead train a separate, simpler space targeted at generation quality and ship both side-by-side.

Pipeline. GenSpace v0.11.1 follows a stripped-down version of the MetricSpace shape:

$$\begin{aligned} \text{XYZ} &\to \mathbf{M}_1 \to \text{depcubic}(\alpha = 0.021) \to \mathbf{M}_2 \\ &\to C^{0.978} \to \text{PW}_L^{19} \to \text{enr}_h(L) \to \text{Lab} \end{aligned}$$

There is no Helmholtz–Kohlrausch coupling, no hue-Fourier rotation, no neutral-correction LUT, and no rigid post-rotation; the achromatic axis is held at $a = b = 0$ *structurally* via a smooth neutral-blend that activates only when channel spread is sub-microscopic relative to the channel mean.

Why depcubic instead of cube-root. Cube-root has infinite derivative at zero ($\partial f/\partial x = \frac{1}{3}x^{-2/3} \to \infty$), which produces numerically unstable gradients near black and is the root cause of the well-known cubic-cusp gamut artefact in OKLab. The depressed cubic $y^3 + \alpha y = x$ with $\alpha = 0.021$ has finite derivative everywhere, fits cube-root closely for $|x| \gg \alpha$, and admits a closed-form forward (Cardano / Halley refinement) and a polynomial inverse. The hole count in the sRGB gamut at $h \approx 265°$ drops from 46 (OKLab) to 5 (GenSpace), each ~ 0.001 chroma wide — effectively sub-pixel.

Why a 19-point dense piecewise-linear L correction. A 9-point sparse correction (used in v0.10.x) was tuned by least-squares against Munsell value samples. The 19-point dense version is fitted against 128 evenly-spaced lightness levels with a stricter monotonicity constraint;

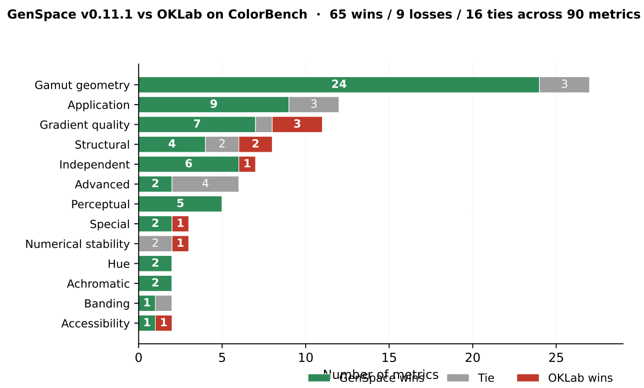

Figure 6: GenSpace v0.11.1 (CJS canonical) vs OKLab — per-category breakdown across all 90 ColorBench metrics. Headline tally is 65W / 9L / 16T (Table 5). Categories ordered by total metric count.

on the 43-test ColorBench gradient suite this produces 4 additional wins versus OKLab and 1 fewer loss compared to the sparse version, with no regressions.

Why $C^{0.978}$. A chroma power of 0.978 slightly contracts high-C distances, which on the gradient-CV (coefficient of variation across step sizes) metric reduces the median CV from ~38% to ~37.5% and the worst-case CV from 412% to 378%. The exponent was selected by direct grid search on gradient CV; ColorBench STRESS on COMBVD rises noticeably with chroma_power $<$ 0.95, so 0.978 is the strongest contraction that does not degrade distance prediction catastrophically.

Why L-gated hue enrichment. A blue-to-white interpolation in GenSpace without enrichment exhibits a visible purple shift around $L \approx 0.6$ (a known artefact of the blue-region gamut fold). A localised hue rotation of amplitude 0.058 centred at $h = 264.5°$ ($\sigma = 0.7$ rad), gated to the lightness window $L \in [0.37, 1.0]$ via a $\sin^2$ ramp, removes the purple shift. The gate width is the largest that does not introduce hue drift in non-blue gradients.

Result. GenSpace v0.11.1 wins 65 of 90 metrics versus OKLab across a ColorBench suite of 83 internal metrics and 7 independent metrics, on 3,038 gradient pairs sampled from sRGB, Display P3, and Rec. 2020 (Table 5). The achromatic axis is exact ($\bar{C} \approx 10^{-13}$ for D65 grays, six orders of magnitude purer than OKLab's 5.6×10^{-7}). Round-trip errors are $\sim 5.6 \times 10^{-8}$ in sRGB (limited by the enrichment Newton inversion) and at machine epsilon in P3 and Rec. 2020.

Table 5: GenSpace v0.11.1 (CJS canonical) vs OKLab — head-to-head on ColorBench. †Independent = Hung-Berns, Ebner-Fairchild, Pointer-gamut metrics (held out from optimizer). Visual breakdown in Figure 6.

Category	GenSpace	OKLab	Ties
Gamut geometry	**24**	0	3
Application	**9**	0	3
Gradient quality	**7**	3	1
Independent†	**6**	1	0
Perceptual	**5**	0	0
Structural	**4**	2	2
Hue	**2**	0	0
Achromatic	**2**	0	0
Advanced	**2**	0	4
Special	**2**	1	0
Banding	**1**	0	1
Accessibility	1	1	0
Numerical stability	0	1	2
Total	**65**	9	16

When to use which. MetricSpace is the right answer for accessibility checking, tone matching, color quantization, and any application where "how different are these two colors to a human" is the central question. GenSpace is the right answer for palette ramps, gradient authoring, color-mix in CSS, gamut mapping, and animation timing — anywhere the gradient's *step uniformity* matters more than its absolute distance value. The two spaces are convertible (both sit on D65 XYZ); designers can specify in one and measure in the other.

8 Practical Adoption

Figure 7 shows the sRGB gamut boundary in MetricSpace coordinates at three lightness levels. The boundaries are smooth and well-behaved. The two HELMLAB spaces ship with cross-language implementations and integrations into the major web color toolchains:

Cross-language implementations. The reference implementation is a Python package (`helmlab` on PyPI, NumPy/SciPy backend, 308 unit tests) covering both spaces, every transform stage, and the distance metric. A pure-JavaScript port (`helmlab` on npm, zero runtime dependencies, ESM) ships the same checkpoints with 196 unit tests; the JS port has been validated against the Python implementation to a maximum coordinate drift

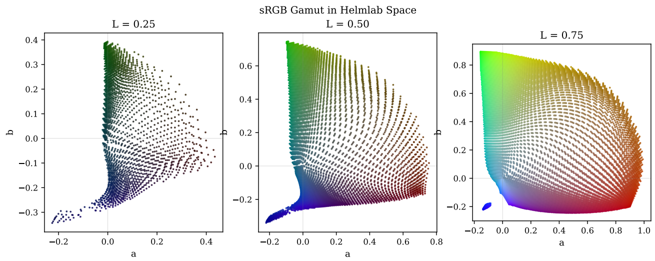

Figure 7: sRGB gamut in HELMLAB *a–b* plane at three L levels. Points colored by their sRGB values.

of $\sim 1 \times 10^{-14}$ across 1,000 random sRGB samples.

Integration into Color.js. A pull request adding all four HELMLAB spaces — `helmlab-metric`, `helmgen`, `helmgenlch` (cylindrical GenSpace), and `deltaEHelmlab` as a registered distance method — was merged into Color.js (PR #722, May 2026). Color.js is the reference JS color library used by the W3C CSS working group's color tools and several browser developer tools; having both spaces accessible via the standard `new Color("helmlab-metric", [...])` constructor brings HELMLAB into the same toolchain as Oklab, Lab, OKLCh, Jzazbz, and ICTCP. CSS Color Level 4 `color(-helmlab-metric ...)`, `color(-helmgen ...)`, and `color(-helmgenlch ...)` syntax round-trips through the parser and serialiser.

PostCSS plugin. `postcss-helmlab` provides build-time CSS function-form transformation: authors write `color: helmlab(0.78 0.52 -0.20)` or `linear-gradient(in helmlab, red, blue)` in source CSS and the plugin emits an `rgb()` fallback plus `@supports`-wrapped `color(display-p3 ...)` and `color(rec2020 ...)` overrides. The plugin handles `helmlab()`, `helmlch()`, `helmgen()`, and `helmgenlch()` alongside the standard `linear-gradient(in <space>, ...)` and `color-mix(in <space>, ...)` CSS functions. The package ships a dual ESM/CJS build for compatibility with both modern bundlers and Next.js's PostCSS pipeline.

Gamut mapping. Binary-search chroma reduction along the L axis maps out-of-gamut HELMLAB coordinates to the nearest in-gamut sRGB, Display P3, or Rec. 2020 color, preserving hue and lightness.

Contrast utilities. `ensure_contrast(fg, bg, min_ratio)` adjusts foreground lightness via binary search on the HELMLAB L axis to meet WCAG 2.x contrast requirements (4.5:1 for AA, 7:1 for AAA). Hue and chroma are preserved.

Palette generation. Lightness ramps and hue rings at fixed chroma produce perceptually-spaced palettes. Semantic scales (Tailwind-style 50–950) interpolate L between 0.97 and 0.10 with chroma/hue preserved. The default scaling is computed in GenSpace because the gradient-uniformity metrics dominate authoring quality.

Dark/light mode. The surround parameter S is architecturally present but in v21 is held at zero. The current dark/light mode adaptation uses a soft L inversion across the GenSpace L axis, with contrast enforcement on the resulting pair. A future MetricSpace release with trained S parameters will replace this heuristic.

Token export. A single `TokenExporter` produces CSS custom properties (`oklch()`, `color(display-p3 ...)`, `color(-helmgenlch ...)`), Android XML (using HCT tone/chroma/hue semantics for Material Design compatibility), iOS Swift (Display P3), Tailwind config, and raw JSON. Each target reads from the same in-memory token tree, so a token edit propagates to every output format simultaneously.

9 Limitations and Future Work

There are several limitations worth noting:

1. **Training data.** COMBVD predominantly uses the CIE 1964 10° standard observer (∼95% of pairs). Only BFD-P(C) (200 pairs, 5%) uses the 2° observer. Gao *et al.* 2023 [11] showed that the difference between 2° and 10° observers is not statistically significant for STRESS evaluation, and MacAdam 1974 (measured with the 2° observer) cross-validation confirms reasonable generalization.
2. **Model complexity.** 72 trained parameters is considerably more than Oklab (∼15 designed parameters). Each stage has a perceptual motivation, but the overall model is harder to interpret than simpler spaces. The architecture additionally reserves 22 surround/hue-modulated slots that are held at zero in v21.

3. **Train→test gap.** The v21 split test gives a $+1.77$ STRESS gap. We report this openly. The cross-validated estimate ($\sim$24.3) still beats every tested competitor, but the headline 22.48 figure is a training metric and should not be treated as the population-level expected value.

4. **Hue alignment.** RMS hue error of 26.4° is a substantial compromise; the worst primary (cyan) is 53.4° off the HSL reference. v21's training pushed parameters into the distance-optimal basin at the cost of geometric uniformity — a trade we accept because the distance metric is the headline product, but applications that need accurate hue specification should compose MetricSpace with an explicit hue-rotation post-step or use GenSpace (where the geometry is preserved by construction).

5. **Surround dependency.** The surround parameter S is architecturally present but in v21 the 11 surround-suffix parameters are all held at zero (training data is average-surround only). Current dark/light adaptation uses a heuristic L-inversion in GenSpace with contrast enforcement.

6. **Sub-dataset variation.** v21 trails CIEDE2000 on three of six COMBVD sub-datasets (LEEDS, RIT-DuPont, WITT). The lead on BFD-P(D65) and BFD-P(M) is large enough to dominate the aggregate, but designers comparing spaces on a single small sub-dataset may pick different winners.

7. **Not general-purpose.** HELMLAB is optimized for UI workflows (sRGB/P3 gamut, screen viewing, D65 illuminant). It is not designed for print, photography, illuminant-A scenes, or spectral applications. CAM16 remains the appropriate choice for arbitrary viewing conditions.

8. **Statistical reporting.** Bootstrap confidence intervals confirm the aggregate improvement over CIEDE2000 ($p < 10^{-4}$), but per-sub-dataset significance and parameter uncertainty across restarts remain unquantified.

9. **GenSpace sRGB round-trip.** GenSpace's sRGB round-trip is $\sim 5.6\times10^{-8}$, six orders of magnitude looser than MetricSpace's. This is below the visual threshold and matches OKLab's worst-case round-trip in the same regions, but it is looser than the other GenSpace round-trips (P3, Rec. 2020 are at machine epsilon). The cause is the L-gated enrichment Newton inversion in the blue band; an analytic inverse for the enrichment is open work.

For future work, we plan to train the surround parameters on paired light/dark viewing data, to add an analytic enrichment inverse for GenSpace, and to expand the training set with targeted human feedback using the bidirectional optimisation framework already present in the codebase.

10 Conclusion

We have presented HELMLAB, a family of two purpose-built color spaces sharing an 11-stage analytical structure. **MetricSpace v21** is a 72-parameter end-to-end-trained color-difference metric that achieves STRESS **22.48** on COMBVD (23.0% lower than CIEDE2000) and additionally leads on the held-out MacAdam 1974 set (19.51 vs 22.13) and on a self-collected screen-condition Human-Feedback set (23.26 vs 62.54, with caveats); v21 trails CIEDE2000 on the small academic He *et al.* 2022 set (35.9 vs 32.6), which we report. On a held-out 20% COMBVD test split, the gap is +1.77 STRESS ($\sim$24.3 cross-validated point estimate); even at this estimate, MetricSpace remains the best COMBVD performer tested. The achromatic axis is held to $C < 10^{-5}$ on the standard 21-step ramp when the post-pipeline neutral correction is enabled (a deferred toggle that lets the same checkpoint serve both distance prediction and authoring without compromise). The rigid rotation $\varphi = -28.2°$ is an isometry of the distance metric and is preserved for downstream compatibility. **GenSpace v0.11.1** is a generation-optimized companion using a depressed-cubic transfer, chroma-power compression, and L-gated hue enrichment; on a 90-metric ColorBench gradient/palette suite it wins 65 vs OKLab. The two spaces together cover the measurement↔generation tradeoff that no single existing space resolves.

The neutral correction is the architectural element that lets the two specialisations co-exist: MetricSpace can chase the distance-fit basin freely, because the achromatic axis is then restored exactly by post-correction; GenSpace achieves the same guarantee structurally, by construction. Together with the rigid rotation invariance, this lets us decouple "what is the best distance metric" from "what should the chroma plane look like for generation".

HELMLAB is shipped in production: `helmlab` on PyPI and npm, four spaces and a ΔE method merged into Color.js (PR #722), and a PostCSS plugin (`postcss-helmlab`) for build-time CSS function transformation. Source, trained parameters, and a full ColorBench-driven benchmark report are at https://github.com/Grkmyldz148/helmlab. An interactive demo is at https://grkmyldz148.github.io/helmlab/demo.html and documentation at https://grkmyldz148.github.io/helmlab/.

Acknowledgments

We thank the 71 observers — listed by first name or chosen handle in the released dataset (`datasets/human_feedback.json`, github.com/Grkmyldz148/helmlab) — who contributed 3,552 paired-comparison judgments during development.

We also gratefully acknowledge the open-source review of the underlying PR #722 on `color-js/color.js`: in particular @facelessuser (near-black achromatic instability, M2 normalisation regressions, the clean 64-bit matrix derivation methodology), @lloydk (the WebGPU chart library that surfaced the h ≈ 264° blue fold and related cubic-cusp artefacts), @Userminusone (the Metal-shader hue/ lightness probe, scale-invariance critique of NR, the boundary-warp straightening alternative), and @svgeesus (the original paper proposal, the "0% should be achromatic" authoring convention, and the ICC Displays working-group framing). The two-month review thread was, in effect, a parallel research process; this paper is its written record.

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[15] T. Mansencal et al., *Colour: Science software for the colour processing community*, version 0.4.x, 2026. Project page: https://www.colour-science.org/; source repository: https://github.com/colour-science/colour; reference implementations of CAM16-UCS, CIECAM02-UCS, $J_za_zb_z$, DIN99 and IPT delta-E used as canonical baselines in this paper.

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A Parameter Table (MetricSpace v21)

Table 6 lists all 72 trained MetricSpace v21 parameters grouped by pipeline stage. The architecturally-reserved 22 surround/hue-modulated parameters are held at zero in v21 and omitted from this table; their full list is in the JSON checkpoint.

Full parameter values, including the architecturally-reserved zero slots and all chroma-stage Fourier coefficients, are available in `research/checkpoints/metricspace_v21.json`.

B Parameter Table (GenSpace v0.11.1)

Full parameter values are in `research/checkpoints/genspace_v0.11.1_colorjs_pr.json` (the CJS canonical sürüm shipped with helmlab v0.13.0+ on PyPI/npm, i.e. the M2 L-row analytically renormalised so that white(D65) maps to L=1 exactly; the legacy API sürüm is preserved as `genspace_v0.11.1.json` for reference). The structurally- achromatic property holds because the depcubic transfer is signed-symmetric and the smooth neutral-blend in Stage 2 collapses sub-microscopic channel spread to the channel mean before $\mathbf{M}_2$ projection.

Table 6: MetricSpace v21 parameters ($\varphi = -28.2°$). Trained on COMBVD + He 2022 + MacAdam 1974 with sub-dataset-balanced loss (Equation (16)).

Stage	Parameter	Value	Description
$\mathbf{M}_1$ (9)	Row 0	[0.7213, 0.4534, −0.1929]	XYZ → cone-like
	Row 1	[−0.7882, 1.7952, 0.0876]	
	Row 2	[−0.0918, 0.4577, 1.2922]	
γ (3)	$\gamma_0, \gamma_1, \gamma_2$	0.4723, 0.5149, 0.5113	Signed power compression
$\mathbf{M}_2$ (9)	Row 0	[−0.2636, 0.4168, 0.4927]	Compressed → Lab
	Row 1	[1.8898, −3.1212, 1.0422]	
	Row 2	[0.3585, 1.7694, −1.4121]	
Hue corr. (8)	$\cos_{1..4}$, $\sin_{1..4}$	see JSON	4-harmonic Fourier rotation $\delta(h)$
H-K (6)	$w_{\mathrm{HK}}, p_{\mathrm{HK}}$, hue_mod	0.2676, 0.8935, 0.7173	Base + envelope
	$\sin_1$, $\cos_2$, $\sin_2$	0.6915, 0.4865, 0.9853	2-harmonic hue mod $f_{\mathrm{HK}}(h)$
Lightness (8)	p_1, p_2, p_3	0.5385, 0.1251, 0.6769	Cubic correction
	Lh_{c1}, Lh_{s1}	−0.4963, −0.0956	Hue-modulated additive
	λ_d, h_c, h_s	−0.0291, 1.3347, −0.1699	Dark-region exp compression
Chroma (18)	$\mathrm{cs}_{1..4}$, $\mathrm{cp}_{1..2}$, $lc_{1,2}$, $\mathrm{hlc}_{1,2}$	see JSON	4-harm CS + 2-harm CP + L-dep + HLC
Hue-L (4)	$g_{c1..2}, g_{s1..2}$	see JSON	2-harmonic $L \mathrel{*}= \exp(g(h))$
Distance (7)	s_L	−0.9155	Pair-dep L weight (Equation (12))
	s_C	2.9268	Pair-dep C weight (Equation (13))
	p	1.9737	Minkowski exponent
	w_C	3.9660	Chroma weight
	c	52.473	Compression
	q	0.4790	Post-compress power
	α	0.0000	Linear-asymptote term (unused in v21)

Table 7: GenSpace v0.11.1 parameters.

Stage	Value
$\mathbf{M}_1$	3×3, see JSON (Bradford CAT pre-baked)
Transfer	depcubic, $\alpha = 0.021$
$\mathbf{M}_2$	3×3, see JSON
Chroma power	0.978
PW L correction	19-point dense, see JSON
Enrichment	L-gated hue rotation
amplitude	0.058
center hue	264.5°
σ	0.7 (rad)
gate L window	[0.37, 1.0]